

Relatório Técnico — Engenharia de Desempenho

API ecommerce-checkout-api | Ferramenta: k6 | Data: 12/06/2026

1. Resumo Executivo (Todos os resultados podem ser encontrados neste [arquivo no github](#))

Endpoint I/O (/checkout/simple): suportou até 300 VUs com 0% de erros e p95 ≈ 290 ms (SLA de 500 ms atendido com folga). Não foi identificado ponto de ruptura dentro dos cenários testados.

Endpoint CPU (/checkout/crypto): colapso a partir de ~150 VUs simultâneos. Com 1.000 VUs, 51,80% das requisições falharam por timeout (p95 = 60 s). Causa raiz: bcrypt.hashSync() síncrono bloqueia o Event Loop do Node.js a cada chamada.

2. Evidências — Resumo dos Testes

Teste (endpoint)	VUs	Requisições	p95	Erros	Resultado
Smoke (/health)	1	81.940	497 µs	0,00%	PASSOU
Load (/simple)	50	6.913	290,24 ms	0,00%	PASSOU
Stress (/crypto)	1.000	7.198	60 s	51,80%	FALHOU
Spike (/simple)	300	17.847	289,97 ms	0,00%	PASSOU

Etapa 1 — Smoke Test (/health, 1 VU, 30 s)

```
■ THRESHOLDS
http_req_duration
  ✓ p(95)<200 p(95)=497.07µs
http_req_failed
  ✓ rate=0 rate=0.00%

■ TOTAL RESULTS
checks_total.....: 163880 5462.59294/s
checks_succeeded...: 100.00% 163880 out of 163880
checks_failed.....: 0.00% 0 out of 163880
✓ status 200
body status 200

HTTP
http_req_duration.....: avg=279.51µs min=154.34µs med=223.98µs max=11.4ms p(90)=334.18µs p(95)=497.07µs
{ expected_response:true }...: avg=279.51µs min=154.34µs med=223.98µs max=11.4ms p(90)=334.18µs p(95)=497.07µs
http_req_failed.....: 0.00% 0 out of 81940
http_reqs.....: 81940 2731.29647/s

EXECUTION
iteration_duration.....: avg=357.87µs min=211.25µs med=288.8µs max=12.55ms p(90)=447.36µs p(95)=652.65µs
iterations.....: 6913 32.825857/s
vus.....: 1 min=1 max=1
vus_max.....: 1 min=1 max=1

NETWORK
data_received.....: 24 MB 789 kb/s
data_sent.....: 6.2 MB 208 kb/s

running (0m30.0s), 0/1 VUs, 81940 complete and 0 interrupted iterations
default ✓ [ 100% ] 1 VUs 30s
Fim : Fri Jun 12 22:06:33 UTC 2026 | Exit code: 0
```

Etapa 2 — Load Test (/checkout/simple, 50 VUs, 3m30s)

```
■ THRESHOLDS
http_req_duration
  ✓ p(95)<500 p(95)=290.24ms
http_req_failed
  ✓ rate=0.01 rate=0.00%

■ TOTAL RESULTS
checks_total.....: 13826 65.651714/s
checks_succeeded...: 100.00% 13826 out of 13826
checks_failed.....: 0.00% 0 out of 13826
✓ status 201
status APPROVED

HTTP
http_req_duration.....: avg=198.89ms min=99.43ms med=108.81ms max=312.52ms p(90)=279.01ms p(95)=290.24ms
{ expected_response:true }...: avg=198.89ms min=99.43ms med=108.81ms max=312.52ms p(90)=279.01ms p(95)=290.24ms
http_req_failed.....: 0.00% 0 out of 6913
http_reqs.....: 6913 32.825857/s

EXECUTION
iteration_duration.....: avg=1.19s min=1.1s med=1.19s max=1.31s p(90)=1.27s p(95)=1.29s
iterations.....: 6913 32.825857/s
vus.....: 2 min=1 max=50
vus_max.....: 50 min=50 max=50

NETWORK
data_received.....: 2.0 MB 9.7 kb/s
data_sent.....: 1.5 MB 7.1 kb/s

running (3m30.6s), 00/50 VUs, 6913 complete and 0 interrupted iterations
default ✓ [ 100% ] 00/50 VUs 3m30s
```

Etapa 3 — Stress Test (/checkout/crypto, até 1.000 VUs, 6m)

```
■ TOTAL RESULTS
checks_total.....: 7198 18.455773/s
checks_succeeded...: 48.10% 3469 out of 7198
checks_failed.....: 51.80% 3729 out of 7198
X status 201
L 40% - ✓ 3469 / X 3729

HTTP
http_req_duration.....: avg=7.48s min=0s med=2.69s max=1m0s p(90)=8.38s p(95)=1m0s
{ expected_response:true }...: avg=9.92s min=89.19ms med=3.2s max=57.4s p(90)=7.14s p(95)=7.66s
http_req_failed.....: 51.80% 3729 out of 7198
http_reqs.....: 7198 18.455773/s

EXECUTION
iteration_duration.....: avg=21.25s min=1.08s med=31s max=1m1s p(90)=31s p(95)=1m1s
iterations.....: 7183 18.417313/s
vus.....: 204 min=2 max=998
vus_max.....: 1000 min=1000 max=1000

NETWORK
data_received.....: 1.1 MB 2.8 kb/s
data_sent.....: 933 kb 2.4 kb/s

running (6m30.0s), 0000/1000 VUs, 7183 complete and 203 interrupted iterations
default ✓ [ 100% ] 0200/1000 VUs 6m0s
Fim : Fri Jun 12 22:29:34 UTC 2026 | Exit code: 0
```

Etapa 4 — Spike Test (/checkout/simple, pico 300 VUs, 1m50s)

```
■ THRESHOLDS
http_req_failed
  ✓ rate<0.05 rate=0.00%

■ TOTAL RESULTS
checks_total.....: 35694 320.891278/s
checks_succeeded...: 100.00% 35694 out of 35694
checks_failed.....: 0.00% 0 out of 35694
✓ status 201
status APPROVED

HTTP
http_req_duration.....: avg=199.89ms min=99.01ms med=199.89ms max=315.25ms p(90)=280.09ms p(95)=289.97ms
{ expected_response:true }...: avg=199.89ms min=99.01ms med=199.89ms max=315.25ms p(90)=280.09ms p(95)=289.97ms
http_req_failed.....: 0.00% 0 out of 17847
http_reqs.....: 17847 160.445639/s

EXECUTION
iteration_duration.....: avg=1.2s min=1.09s med=1.2s max=1.31s p(90)=1.28s p(95)=1.29s
iterations.....: 17847 160.445639/s
vus.....: 2 min=1 max=300
vus_max.....: 300 min=300 max=300

NETWORK
data_received.....: 5.3 MB 48 kb/s
data_sent.....: 4.0 MB 36 kb/s

running (1m51.2s), 000/300 VUs, 17847 complete and 0 interrupted iterations
default ✓ [ 100% ] 000/300 VUs 1m50s
Fim : Fri Jun 12 22:41:26 UTC 2026 | Exit code: 0
```

3. Análise de Estresse — Ponto de Ruptura

Primeiros timeouts registrados às 22:24:34 UTC com ~150 VUs ativos (1m28s de execução, ramp-up 0→200 VUs em 2 min). Causa: bcrypt.hashSync() síncrono satura o Event Loop do Node.js com requisições concorrentes.

Faixa de VUs	Comportamento observado
0 – 148 VUs	Estável — requisições respondidas normalmente
~150 VUs	Primeiros timeouts — PONTO DE RUPTURA identificado
200 – 500 VUs	Degradação severa — latência média > 7,48 s
500 – 1.000 VUs	Colapso total — 51,80% de falhas, p95 = 60 s (timeout máximo)